Lorena Marcos

Machine Learning & AI | Applied Data Science Student

Madrid, Spain · Available from February 2027, 20–30 h/week (afternoons)
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SUMMARY

Applied Data Science undergraduate at Universidad Complutense de Madrid, focused on machine learning, AI engineering and building end-to-end software products. Hands-on experience with Python, SQL, predictive modeling and statistical analysis, plus a full-stack platform whose backend is deployed to production for a real event pilot. Seeking a paid, part-time ML/AI role from February 2027, with the goal of moving to full-time after graduation.

EDUCATION

BSc in Applied Data Science — Universidad Complutense de Madrid 2023 – Present

- Relevant coursework: Machine Learning, Supervised & Unsupervised Learning, Data Mining, Time Series, Statistical Inference, Linear Algebra, Discrete Mathematics.

PROJECTS

Ride-Sharing Cancellation Prediction — Machine Learning 2025

XGBoost · R · tidyverse · dbscan · Quarto

- Built an XGBoost classification model to predict trip cancellations on ~150,000 ride-sharing records.
- Engineered temporal, geographic and user-behaviour features; full pipeline from EDA and cleaning to training and evaluation.
- Handled structural missingness (MNAR) and duplicate IDs; detected anomalies with IQR and Local Outlier Factor.

github.com/Lorenadata/ncr-ride-sharing-eda

RaceTrack — Sports Event Management Platform (independent project) 2026 – Present

Python · FastAPI · SQLAlchemy · Alembic · PostgreSQL/PostGIS (Supabase) · Redis · React PWA · Stripe · Railway

- Building a full-stack platform for popular races and charity walks: participant registration, payments, maps and push notifications.
- Designed and deployed the REST API backend to production; database migrations, JWT authentication and Stripe webhooks.
- QR-based bib verification through a dedicated API endpoint; geospatial data with PostGIS and MapLibre.
- Pilot with a real charity march of 12,000–15,000 participants, working directly with the event organizer.

Fiscalis — Database for a Tax Management SaaS 2025

SQL · PostgreSQL · Relational design

- Designed the relational model of an MVP for self-employed users (invoices, clients, fees) with SQL queries for treasury, tax obligations and payment calendar.

github.com/Lorenadata/fiscalis-db

Income Inequality & Child Poverty in Spain — Official Microdata 2025

R · dplyr · Hmisc · ggplot2 · Quarto · INE ECV 2024

- Survey-weighted analysis (RB050) of tenants vs. landlords and child poverty (AROPE): weighted percentiles and P90/P10 ratio; published as reproducible reports.

lorenadata.github.io/ECV-rental-market

EXPERIENCE

Solar Energy Hackathon — Predictive Pricing Model 2024

- Built a photovoltaic energy price prediction model in a multidisciplinary team in under 48 hours (Python, feature engineering); presented to a technical panel.

SKILLS

Programming	Python (pandas, NumPy), SQL, R (tidyverse, ggplot2, Quarto)
Machine Learning	scikit-learn, XGBoost, statsmodels; classification, regression (GLM), supervised & unsupervised learning
Statistics	Statistical inference, survey weighting, outlier detection (LOF, IQR), missing-data analysis, time series
AI	Anthropic Claude API for workflow automation
Backend & Data	FastAPI, REST APIs, PostgreSQL, PostGIS, MongoDB, SQLAlchemy, Alembic, Redis
Tools	Git, GitHub, Railway, Jupyter, React
Languages	Spanish (native), English (B2)